Samuel Paul Lum

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EDUCATION

VIRGINIA TECH
Bachelor of Science in Mechanical Engineering *Expected May 2028*
GPA: 2.887

TECHNICAL SKILLS

CAD & Design: Siemens NX, Fusion 360, Engineering Drawings, GD&T, Product Design
Programming & Analysis: MATLAB (Numerical Simulation), Python, Arduino, Excel, Git, Statistical Analysis
Manufacturing & Fabrication: CNC Machining, G-Code, Laser Cutting, 3D Printing, Welding, Sand Casting
Certifications: MATLAB Onramp and MATLAB Desktop Tools & Troubleshooting Scripts (MathWorks),
Six Sigma White Belt, OSHA-10, SOLIDWORKS Associate (in progress), Google Data Analytics (in progress)

WORK EXPERIENCE

Westwood Professional Services | Staunton, Virginia
Survey/Civil Engineering Intern *May 2026 – Aug 2026*

- Process total station and GPS survey data using Carlson Survey and AutoCAD Civil 3D
- Draft survey linework, surface models, grading plans, and plats for residential and commercial land development
- Perform field surveys, including as-built and improvement surveys, to collect and verify accurate site data
- Collaborate with engineers and survey crews by balancing office drafting and data processing with field data collection

Moffett Paving and Excavating | Staunton, Virginia
Construction Crew Member *May 2024 - Aug 2025*

- Supported residential and municipal paving, excavation, grading, and site preparation projects
- Operated rollers and backhoes while aiding with grade verification using laser levels to support accurate earthwork
- Utilized construction equipment and tools, including plate compactors, heavy-duty saws, and impact tools

PROJECTS

Mazda MX-5 Miata Maintenance & Electrical Repair (Personal Automotive Project) *July 2026*

- Replaced the catalytic converter, working under the vehicle to remove and realign the exhaust connections
- Traced and repaired headlight wiring, verified the headlight assemblies, and inspected suspension for wear

Smart Trash Bag Attachment (Sophomore Virginia Tech Engineering Design Project) *Jan 2026 - May 2026*

- Led the CAD design of a mechanical attachment intended to automate the sealing and removal of trash bags
- Designed components in Siemens NX and prepared models for additive manufacturing using a Bambu Lab 3D printer
- Fabricated and assembled a proof-of-concept prototype through multiple design iterations
- Collaborated with a multidisciplinary engineering team to design and identify opportunities for future improvement

Smart Crosswalk Safety System (Freshman Virginia Tech Engineering Design Project) *Jan 2025 - May 2025*

- Served as the primary CAD designer for a smart pedestrian crossing system intended to improve crosswalk safety
- Designed structural components in Fusion 360 and assisted in constructing and testing the prototype
- Integrated infrared sensors and ground lighting to demonstrate the system's safety concept
- Collaborated with a multidisciplinary engineering team throughout design, fabrication, and testing

EXTRACURRICULARS

American Society of Mechanical Engineers (ASME) Member *Sept 2025 – Present*

Virginia Tech Men's Soccer | Blacksburg, Virginia *Aug 2025 – Present*
Student Manager

- Coordinate equipment logistics for practices, competitions, and team travel for the 29 players
- Manage setup, inventory, and transportation of training equipment to support daily team operations
- Assist coaches with practice and game-day preparation for a Division I NCAA program